



GC1015

FROM KPIS TO DEFENSIBLE DATA STORIES

Statistics, variables, data grain, and chart critique



Concepts first, guided
practice after the break

TODAY'S DECISION-TO-STORY PATH

Concepts first, guided practice after the break

1

Key Point 1

- Decision Frame and Running-Case Brief

2

Key Point 2

- Operational **KPI** Definitions and Statistical Meaning

3

Key Point 3

- Variables, Measurement Scales, Observations, and Grain



Slides hand off to the lab, the lab hands back

TODAY'S LOOP: SLIDES AND LAB

Every section ends in your notebook and workbook; every practice returns you here

1

Learn, then practise

- After each section, complete six minutes of practice. Use the notebook and workbook for a fresh example and the *MetroMeal* example.

2

Break after Section 4

- Return for Section 5, then the 15-minute guided build. Watch the first step and complete the remaining steps in Excel.

3

Then the studio and the close

- Use your checked evidence in the group studio, then complete the synthesis and exit ticket. Keep the lab notebook and Excel workbook open.



Chart Selection and
Honest Statistical
Critique

TODAY'S DECISION-TO-STORY PATH | CONTINUED 2

Concepts first, guided practice after the break

1

Key Point 1

- Chart Selection and Honest Statistical Critique

2

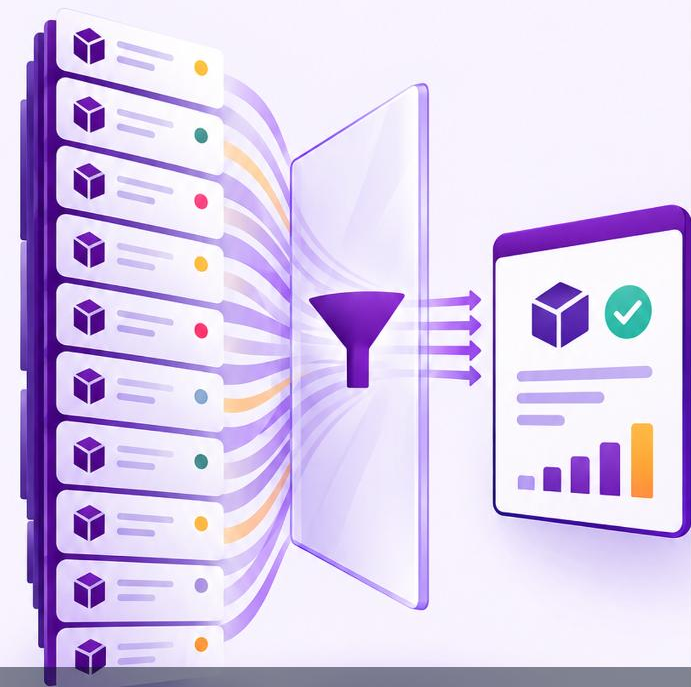
Key Point 2

- Bounded Story, Validation, and Bridge to Descriptive Evidence

3

After the break

- Guided Excel build and MetroMeal feedback studio



Analysts connect
evidence to decisions,
not just data to
graphics.

WHY THIS MATTERS

Software can draw a chart, but it cannot choose an honest story

1

Key point

- Analysts connect evidence to decisions, not just data to graphics.
- A wrong **denominator** (the eligible total), mixed **grain** (what one row represents), misleading title, or distorted axis can turn correct arithmetic into a false impression.

2

Key point

- Professional value comes from defining the measure, checking where the data came from, stating limits, and proposing the next check.



METROMEAL DELIVERY

One staffing decision, four descriptive weeks, and no license to overclaim

METROMEAL DELIVERY | CASE BRIEF

One staffing decision, four descriptive weeks, and no license to overclaim

1

Your role and decision

- Regional operations analyst supporting MetroMeal Delivery's operations director.
- Decide whether to investigate weekend staffing before the next scheduling cycle. Use the delivery **KPI**, target, complaint guardrail and four-week evidence; separate description from explanation.

2

Evidence and scope

- **Use orders_story.csv. Observed window:** August 3 through August 30, 2026. Decision horizon: the next MetroMeal staffing cycle. Latest on-time rate: 78.4%, below the 85.0% target.

Regional operations
analyst supporting
MetroMeal Delivery's
operations director.

METROMEAL DELIVERY | CASE DETAILS 2

One staffing decision, four descriptive weeks, and no license to overclaim



Latest complaint rate:
2.8%, within the 3.0%
maximum guardrail.

1

Evidence and scope

- **Latest complaint rate:** 2.8%, within the 3.0% maximum guardrail. Latest week: 500 eligible deliveries, including 392 on-time deliveries and 108 late deliveries.
- **Reframed question:** investigate staffing given an 85.0% target and 3.0% complaint guardrail.

2

Expected output

- Can you trace a business story from decision through **KPI**, grain, chart, claim, and validation? State the evidence and its limits before recommending an action.

SECTION

1. DECISION FRAME AND RUNNING-CASE BRIEF

Translate a chart request into a decision the evidence can inform



DECISION-FIRST ANALYSIS

Define the managerial choice before selecting the measure or chart

1

Definition and rule

- **Decision-first analysis** defines the stakeholder, plausible choice, action horizon, outcome, comparison, and required evidence before analysis begins.
- Decision frame = stakeholder + choice + action horizon + outcome + comparison + evidence



Observed window:
August 3 through
August 30, 2026.

DECISION-FIRST ANALYSIS | CONTINUED 2

Define the managerial choice before selecting the measure or chart

1

MetroMeal example

- **Observed window:** August 3 through August 30, 2026. Action horizon: before the next staffing cycle. Choice: investigate weekend staffing or continue current scheduling.

2

Limit and check

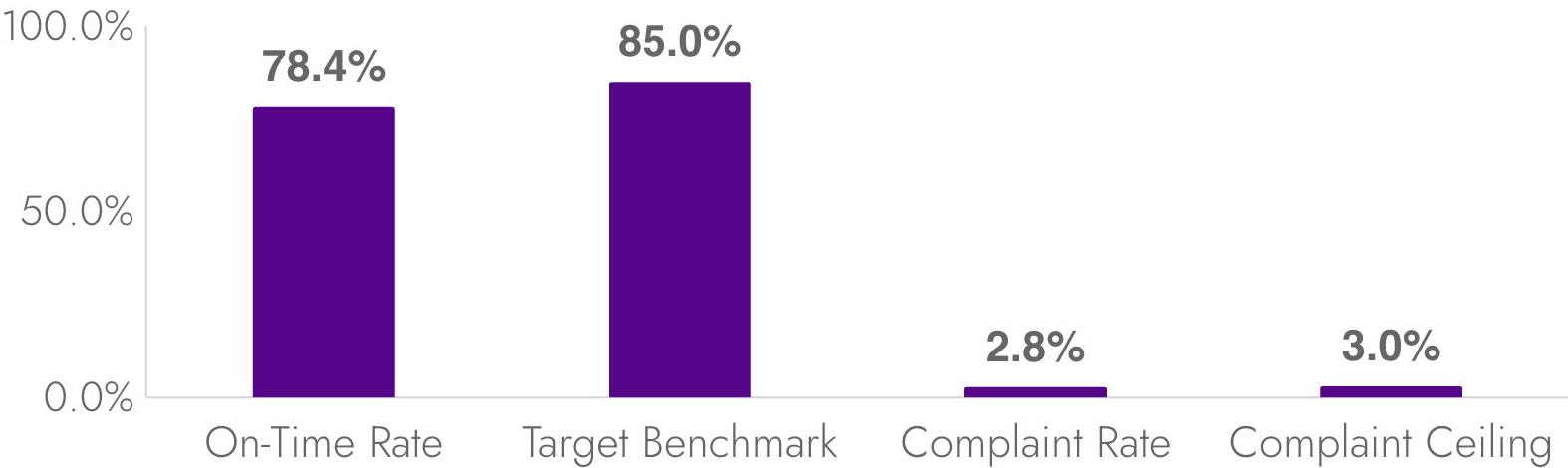
- These records describe what happened. They can support an investigation, but they cannot prove that staffing caused the result.



Three related terms with
different managerial
roles

KPI, TARGET, AND GUARDRAIL

Three related terms with different managerial roles



MetroMeal achieved a 78.4% on-time delivery rate against an 85.0% target while holding complaints to 2.8% against a 3.0% ceiling.

eligible delivery orders in Week 4 | August 24 to August 30, 2026 | Unit: percent of eligible orders



A Key Performance Indicator (KPI) monitors an objective.

KPI, TARGET, AND GUARDRAIL

Three related terms with different managerial roles

1

Definition and rule

- A **Key Performance Indicator (KPI)** monitors an objective. A **target** is the desired benchmark. A **guardrail** limits harm to another important outcome.



KPI: on-time proportion
= $392 / 500 = 78.4\%$.

KPI, TARGET, AND GUARDRAIL | CONTINUED 2

Three related terms with different managerial roles

1

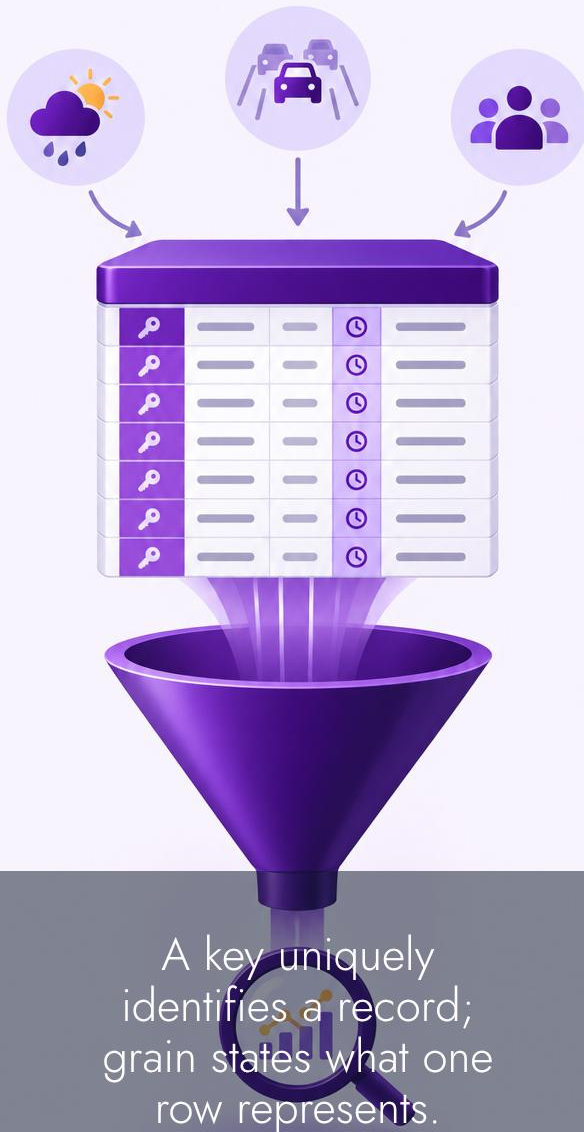
Worked example and meaning

- **KPI:** on-time proportion = $392 / 500 = 78.4\%$. Target: at least 85.0%. Guardrail: complaint proportion = $14 / 500 = 2.8\%$, with a maximum of 3.0%.
- MetroMeal missed the target by 6.6 percentage points but remained within the complaint guardrail, so investigation is appropriate.

2

Limit and check

- The 85.0% target is a management benchmark, not the **KPI** or a value estimated from these records.



METROMEAL'S BOUNDED QUESTION

Descriptive records identify a target miss, not its cause

1

Evidence boundary

- A **key** uniquely identifies a record; **grain** states what one row represents.
- MetroMeal uses order_id as a candidate **key**. Each of the 2,000 rows represents one eligible completed delivery from August 3 through August 30, 2026.



Should the director investigate weekend staffing before the next delivery cycle?

METROMEAL'S BOUNDED QUESTION | CONTINUED 2

Descriptive records identify a target miss, not its cause

1

Worked example and decision

- Week 4 had 392 on-time and 108 late deliveries: $392 / 500 = 78.4\%$.
- Should the director investigate weekend staffing before the next delivery cycle?

2

Limit and check

- The records establish a target miss, but descriptive data do not prove that staffing caused it.

AI SEARCH & SHARE



Connect a verified
metric to a measurable
consequence

ASK AI: A REAL KPI IN PRACTICE

Connect a verified metric to a measurable consequence

1

Ask AI

- "Investigate Delta Air Lines. Find one operational **KPI** and its stated target or benchmark in an official report dated 2023 to 2026.
- Identify a measurable consequence of missing it in money, time, flights, customers, or risk. State when a target is not publicly disclosed."

2

Share

- Name the **KPI**, benchmark, consequence, and one uncertainty.
- Verify the key number with a dated credible source outside the **AI** answer. Do not upload private data.

RETRIEVAL CHECK

Answer before viewing the debrief

- **Retrieval challenge**

Question Which term names the monitored measure, which names management's benchmark, and which protects against secondary harm? Take 30 seconds. Name the three terms and map each one to MetroMeal before reviewing the answer.

- **Retrieval challenge**

Next topic: Operational KPI Definitions and Statistical Meaning.



Excel first, SPSS second

CLASS PRACTICE 1: FRAME THE DECISION

6 minutes in the lab, then back here (slide 20)

1

Open

- Open Session 02 Practice at “Class practice 1: What decision can the **KPI** inform?” Excel cells: **B6:B14, B10, B11, B12, B13, B14**.

2

Do, Excel then SPSS

- Use the lab prompts to frame MetroMeal’s decision, **KPI**, 85.0% target, 3.0% complaint guardrail, supported claim, and unsupported claim.
- **SPSS block:** “Block **H1**: decision, **KPI**, target, and guardrail”.

3

Bring back

- Bring back one bounded decision question, one supported descriptive claim, and one causal claim to remove. Timer: 6 minutes.

SECTION

2. OPERATIONAL KPI DEFINITIONS AND STATISTICAL MEANING

Specify, calculate, interpret, and aggregate a reproducible KPI

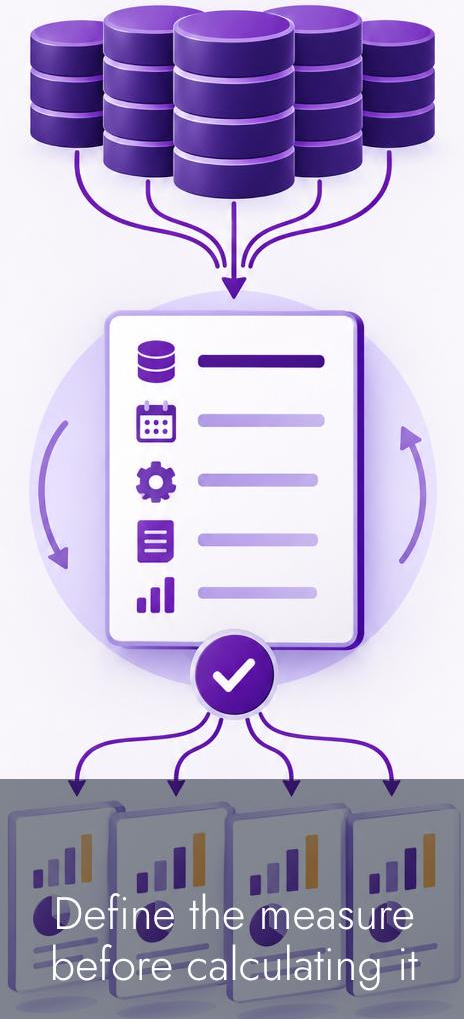
REPRODUCIBLE KPI SPECIFICATION

Define the measure before calculating it

1

Definition and rule

- A **KPI specification** defines the measure name, business purpose, unit, population, reporting period, row grain, eligibility rules, formula, and benchmark.





One row represents one delivery; canceled and test orders are excluded.

REPRODUCIBLE KPI SPECIFICATION | CONTINUED 2

Define the measure before calculating it

1

Worked example and meaning

- MetroMeal defines its weekly on-time rate over 500 completed deliveries from August 24 to August 30, 2026. One row represents one delivery; canceled and test orders are excluded.

2

Limit and check

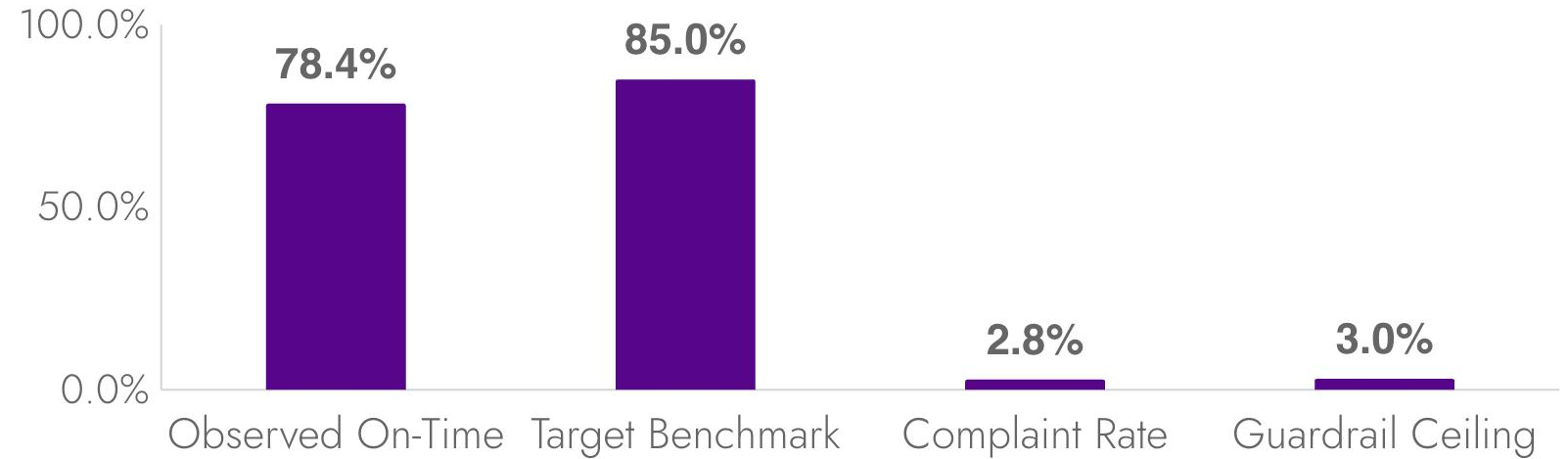
- Changing an eligibility filter or reporting window creates a new metric version and can produce an artificial performance shift.



Preserve the
denominator and label
the difference correctly

COUNT, PROPORTION, AND TARGET GAP

Preserve the denominator and label the difference correctly



MetroMeal achieved a 78.4% on-time rate against an 85.0% target (-6.6 percentage points), while keeping complaints at 2.8%, within the 3.0% guardrail.

eligible completed deliveries | August 24-30, 2026 | Unit: percent of eligible orders



A count is the absolute frequency of qualifying events.

COUNT, PROPORTION, AND TARGET GAP

Preserve the denominator and label the difference correctly

1

Definition and rules

- A **count** is the absolute frequency of qualifying events. A **proportion** divides that **count** by the eligible total.
- On-Time Rate = (On-Time Deliveries) / (Eligible Completed Deliveries) × 100%

2

Definition and rules

- Target Gap = Observed Rate - Target Rate



MetroMeal recorded
 $(392) / (500) = 78.4\%$.

COUNT, PROPORTION, AND TARGET GAP | CONTINUED 2

Preserve the denominator and label the difference correctly

1

Worked example and meaning

- MetroMeal recorded $(392) / (500) = 78.4\%$. The target gap is $78.4\% - 85.0\% = -6.6$ percentage points (below target). Complaints were $(14) / (500) = 2.8\%$, within the 3.0% guardrail.

2

Limit and check

- Subtracting two percentages produces **percentage points**, not a percent change.

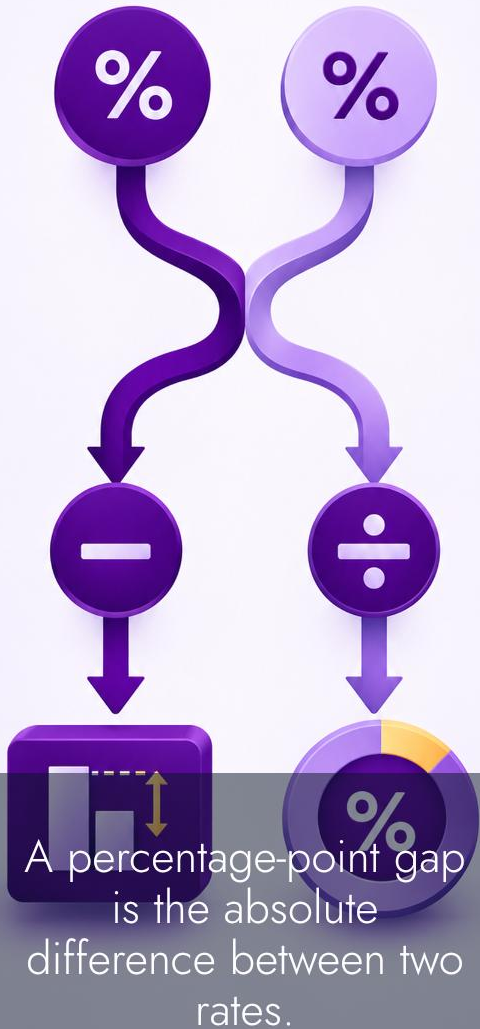
PERCENTAGE POINTS VS PERCENT CHANGE

Subtracting rates differs from dividing by a baseline

1

Definition and rules

- A percentage-point gap is the absolute difference between two rates. Percent change compares that difference against a baseline.
- $\text{Percent change} = (\text{new rate} - \text{baseline rate}) / \text{baseline rate} \times 100\%$.





$78.4\% - 85.0\% = -6.6$
percentage points.

PERCENTAGE POINTS VS PERCENT CHANGE | CONTINUED 2

Subtracting rates differs from dividing by a baseline

1

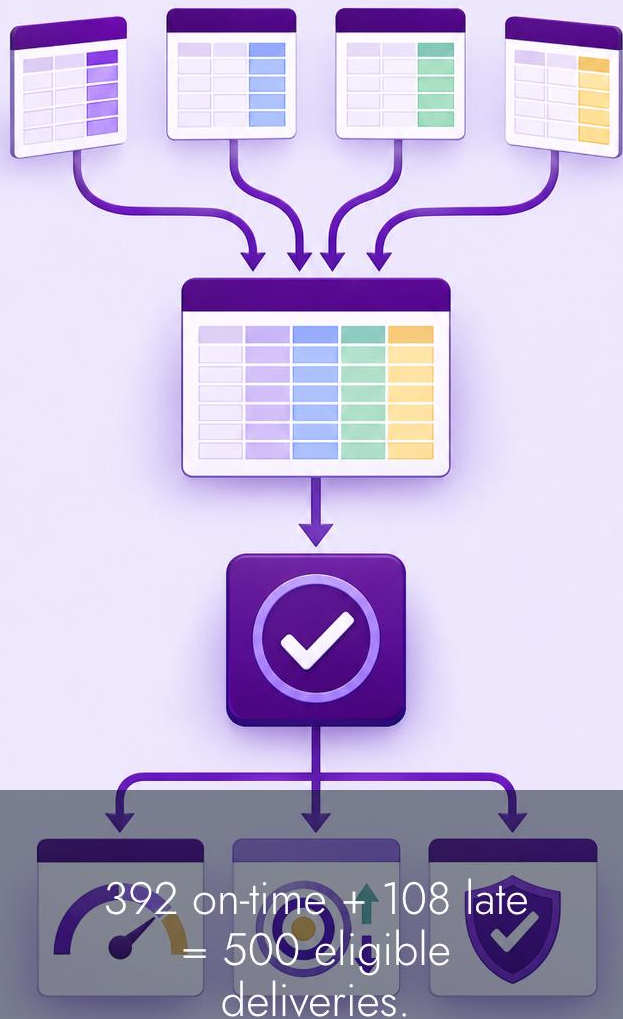
Worked example and meaning

- $78.4\% - 85.0\% = -6.6$ percentage points. Relative to the target baseline, $-6.6 / 85.0 \times 100\% = -7.76\%$.

2

Limit and check

- Negative 6.6 percentage points is the target gap. Negative 7.76% is the relative shortfall; do not mix the units.



REPRODUCE THE KPI IN EXCEL 365

Inspect the inputs before trusting the percentage

1

Reconcile the counts

- Use **Slide_Practice cells B21:B27** for the latest-week counts, rate, target gap and complaint rate. Check that 392 on-time plus 108 late deliveries equals 500 eligible deliveries.

2

Calculate separately

- $392 / 500 = 78.4\%$; $78.4\% - 85.0\% = -6.6$ percentage points; $14 / 500 = 2.8\%$.

3

Recovery check

- If the count check returns **FALSE**, stop and inspect filters, missing records, or duplicate rows. Predict first: Is $392 / 500$ above or below 80%?



The mean is 12.4 minutes and the median is 7.0 minutes.

CENTER AND POSITION IN A KPI MAP

Different summaries answer different questions about delay severity

Measure	Statistical role	Delay (minutes)
Mean	Center: arithmetic average	12.4
Median	Center: ordered midpoint	7
90th percentile	Position: PERCENTILE.INC	31

The mean is 12.4 minutes and the median is 7.0 minutes. The 90th percentile is 31.0 minutes under Excel PERCENTILE.INC. Ties and interpolation affect the percentage at or below a percentile cutoff.

108 late eligible completed deliveries | August 24 to August 30, 2026 | Unit: minutes



The mean and median describe center; a percentile describes relative position.

CENTER AND POSITION IN A KPI MAP

Different summaries answer different questions about delay severity

1

Definition and rules

- The mean and median describe center; a percentile describes relative position. For an even count, the median averages the two middle sorted values.
- State the variable, unit, subset, period, and convention. Mean = total eligible minutes / eligible observations.



Different summaries
answer different
questions about delay
severity

CENTER AND POSITION IN A KPI MAP | CONTINUED 2

Different summaries answer different questions about delay severity

1

Worked example and meaning

- **Among 108 late deliveries:** mean delay = 12.4 minutes, median delay = 7.0 minutes, and the Excel **PERCENTILE.INC** 90th **percentile** = 31.0 minutes.

2

Limit and check

- A mean above the median suggests an uneven upper tail, but requires visual distribution checks before claiming skew.



Combine numerators
and denominators
before calculating the
rate

AGGREGATION FROM COUNTS

Combine numerators and denominators before calculating the rate

Week	On-Time Deliveries	Eligible Deliveries	Weekly Rate
Week 1	420	500	84.0%
Week 2	438	500	87.6%
Week 3	425	500	85.0%
Week 4	392	500	78.4%
Total	1,675	2,000	83.75%

Pooling counts yields an overall rate of 83.75%, matching the simple average only because all weekly denominators equal 500.

eligible customer orders | August 3 to August 30, 2026 | Unit: deliveries and percent on time



An overall rate pools qualifying counts and eligible counts before dividing.

AGGREGATION FROM COUNTS

Combine numerators and denominators before calculating the rate

1

Definition and rule

- An overall rate pools qualifying counts and eligible counts before dividing.
- Overall rate = $\Sigma \text{ on-time deliveries} / \Sigma \text{ eligible deliveries} \times 100\%$



On-time deliveries: $420 + 438 + 425 + 392 = 1,675$.

AGGREGATION FROM COUNTS | CONTINUED 2

Combine numerators and denominators before calculating the rate

1

Worked example and meaning

- **On-time deliveries:** $420 + 438 + 425 + 392 = 1,675$. Eligible deliveries: $500 \times 4 = 2,000$.
- **Four-week rate:** $1,675 / 2,000 = 83.75\%$, falling 1.25 points below the 85.0% target.

2

Limit and check

- Averaging weekly percentages matches here only because every week has exactly 500 deliveries. When volumes vary, always pool raw counts.

AI SEARCH & SHARE



3 minutes to investigate,
2 minutes to share

ASK AI: VERIFY SOUTHWEST'S CANCELLATION RATE

3 minutes to investigate, 2 minutes to share

1

Ask AI

- “Southwest Airlines reported a 14.6% cancellation rate for December 2022. Explain the numerator, denominator, reporting period, and operational meaning.
- Connect this rate to the 16,900 canceled flights, two million stranded passengers, and the \$140 million civil penalty reported by the United States Department of Transportation on December 18, 2023.



Cite official dated sources and state one uncertainty."

ASK AI: VERIFY SOUTHWEST'S CANCELLATION RATE | CONTINUED 2

3 minutes to investigate, 2 minutes to share

1

Ask AI

- Evaluate evidence for and against treating the monthly rate and holiday disruption as identical analytical windows. Cite official dated sources and state one uncertainty."

2

Share

- State the cancellation rate, denominator, business consequence, dated external source, period alignment, and one remaining uncertainty. Do not upload private data.

RETRIEVAL CHECK

Explain the unit before viewing the debrief

- **Question**

Why is MetroMeal's Week 4 target gap negative 6.6 percentage points rather than a negative 6.6% drop?

- **Student Task**

Take 30 seconds. Write down the difference formula, calculated gap, correct unit, and the formula required for relative percent change.

- **Next Topic**

Variables, Measurement Scales, Observations, and Grain.



Excel first, SPSS second

CLASS PRACTICE 2: DEFINE AND CHECK THE MEASURE

6 minutes in the lab, then back here (slide 39)

1

Open

- Open Session 02 Practice at “Class practice 2: Is the **KPI** calculation reproducible?” Excel cells: **B21, B22, B23, B24:B25, B26:B27, B28: B29**.

2

Do, Excel then SPSS

- Use the lab prompts to verify the latest 392 of 500, 78.4% rate, -6.6 percentage-point gap, 2.8% complaint rate, and four-week 1,675 of 2,000 aggregate.
- **SPSS block:** “Block **H2**: weekly and aggregate **KPI** calculations”.

3

Bring back

- Bring back a checked **KPI** formula with units, visible numerator and denominator, target gap, and one limitation. Timer: 6 minutes.

SECTION

3. VARIABLES, MEASUREMENT SCALES, OBSERVATIONS, AND GRAIN

Determine what each field means and what each row represents



Columns record
characteristics; rows
represent complete
cases

VARIABLES AND OBSERVATIONS

Columns record characteristics; rows represent complete cases

order_id	item_count	delivery_minutes	satisfaction	complaint
10482	3	54	2	No

The columns are five variables, while the complete row is one delivery observation; the numeric-looking order ID remains a label.

one eligible completed delivery | August 8, 2026 | Unit: recorded delivery attributes



A variable is a recorded characteristic.

VARIABLES AND OBSERVATIONS

Columns record characteristics; rows represent complete cases

1

Variables and observations

- A variable is a recorded characteristic. An observation is all recorded values for one entity or event at the table's grain.

2

Rule and example

- Variables are columns; observations are rows. Order 10482 is one eligible completed delivery recording 3 items, 54 delivery minutes, satisfaction category 2, and complaint status No.

3

Check

- Digits do not make a field numerical. Averaging order_id has no business meaning.

DATA TYPE AND MEASUREMENT SCALE

Classify each field on two separate dimensions

1

Two classifications

- Data type asks category or number; numerical variables may be discrete counts or continuous measurements.
- Measurement scale asks what comparisons are valid: nominal = labels, ordinal = ordered categories, interval = equal gaps without a true zero, ratio = equal gaps with a true zero.



Classify each field on
two separate
dimensions



Classify by meaning,
not cell format.

DATA TYPE AND MEASUREMENT SCALE | CONTINUED 2

Classify each field on two separate dimensions

1

MetroMeal

- order_id is categorical nominal; item_count is numerical discrete and ratio; satisfaction 1-5 is categorical ordinal; delivery_minutes is numerical continuous and ratio.

2

Check

- Classify by meaning, not cell format. Nominal and discrete answer different questions.



WHY NUMBERS CAN BE NOMINAL OR ORDINAL

Storage type does not dictate arithmetic meaning

1

Storage versus meaning

- Software stores digits as numeric data types, but business meaning dictates valid arithmetic operations.

2

MetroMeal scale check

- order_id 10482 is nominal, satisfaction = 2 is ordinal, and delivery_minutes = 54 is ratio.

3

Validity rule

- Excel executes **AVERAGE**(order_id) without an error, but the resulting **average** has zero statistical meaning.

ROW GRAIN AND SUPPORTED CLAIMS

State what one row means before summarizing the data



State what one row means before summarizing the data

Table	One row represents	Rows	Supports
Raw	eligible completed delivery	2000	delivery-level checks
Weekly	week	4	weekly rate comparisons

Aggregation reduces 2,000 delivery rows to four weekly rows and removes individual delivery detail.

2,000 eligible completed deliveries | August 3 through August 30, 2026 | Unit: rows at the stated grain



State what one row
means before
summarizing the data

ROW GRAIN AND SUPPORTED CLAIMS

State what one row means before summarizing the data

1

Row grain definition

- Grain specifies what one row represents, defining entity level, detail level, and time boundary.

2

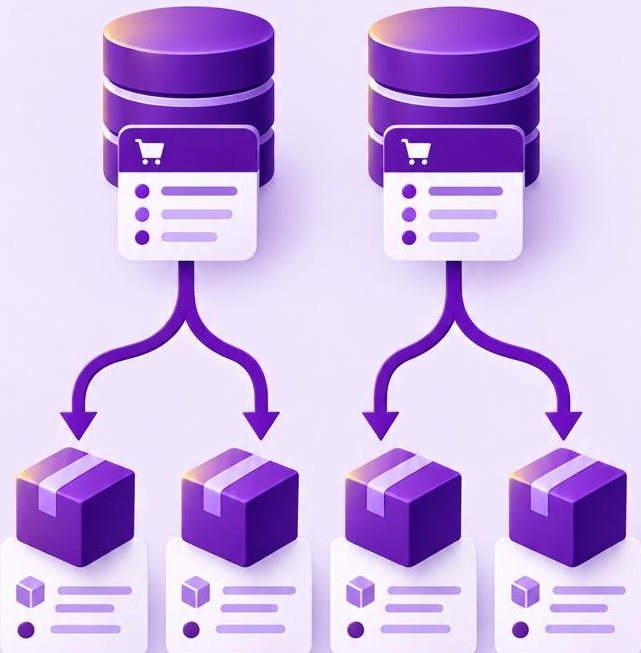
MetroMeal comparison

- Raw data has 2,000 completed delivery rows; the summary table has 4 weekly rows.

3

Analytical limits

- Delivery rows support order-level checks, while weekly rows support trend analysis; aggregation discards transaction detail.



MIXED-GRAIN JOIN RISK

Repeated entities can silently change analytical weight

1

Join multiplication

- Merging tables at different grains duplicates parent rows and distorts calculated performance metrics.

2

MetroMeal order join

- Joining Orders to Items expands two orders into four rows because Order A contains three items.

3

Audit check

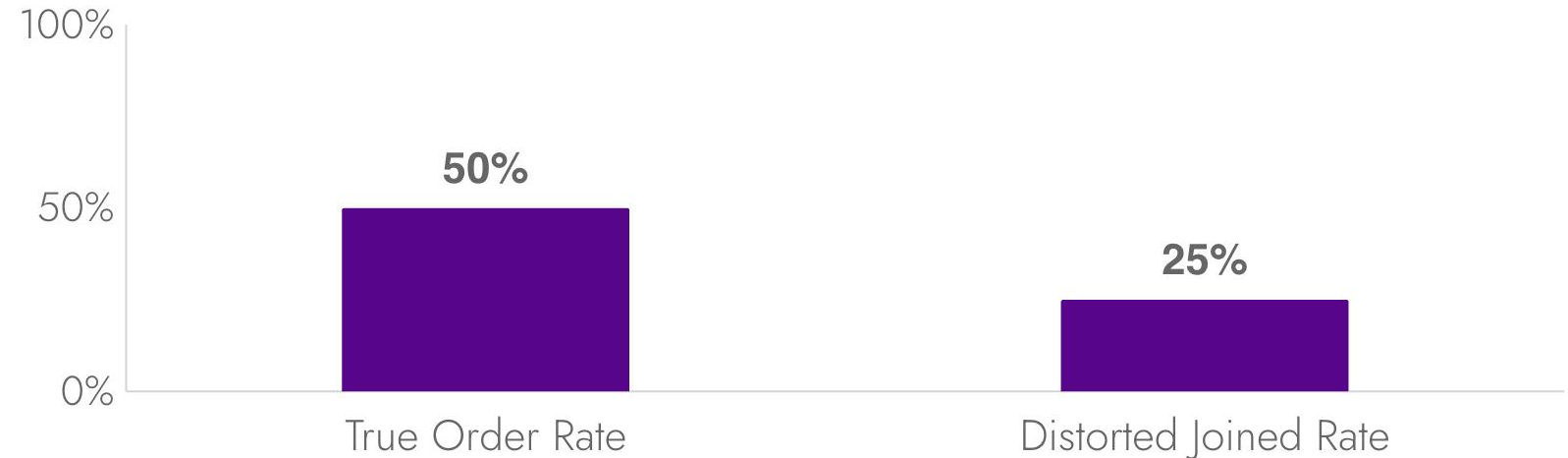
- Validate table relationships and row counts before calculating order KPIs, or aggregate child items first.



How multi-item orders
distort delivery-level
KPIs

VISUALIZING THE JOIN FAN TRAP

How multi-item orders distort delivery-level KPIs



Order A has 3 items (late) and Order B has 1 item (on-time). Naive row aggregation cuts reported on-time performance from 50.0% to 25.0% without any change in operational delivery.

two orders in a separate illustrative join example | separate teaching example, not the four-week KPI dataset | Unit: percent on-time



The true on-time rate is
 $1 / 2 = 50.0\%$.

VISUALIZING THE JOIN FAN TRAP

How multi-item orders distort delivery-level KPIs

1

Worked Example

- At order grain, Order A is late and Order B is on time. The true on-time rate is $1 / 2 = 50.0\%$.

2

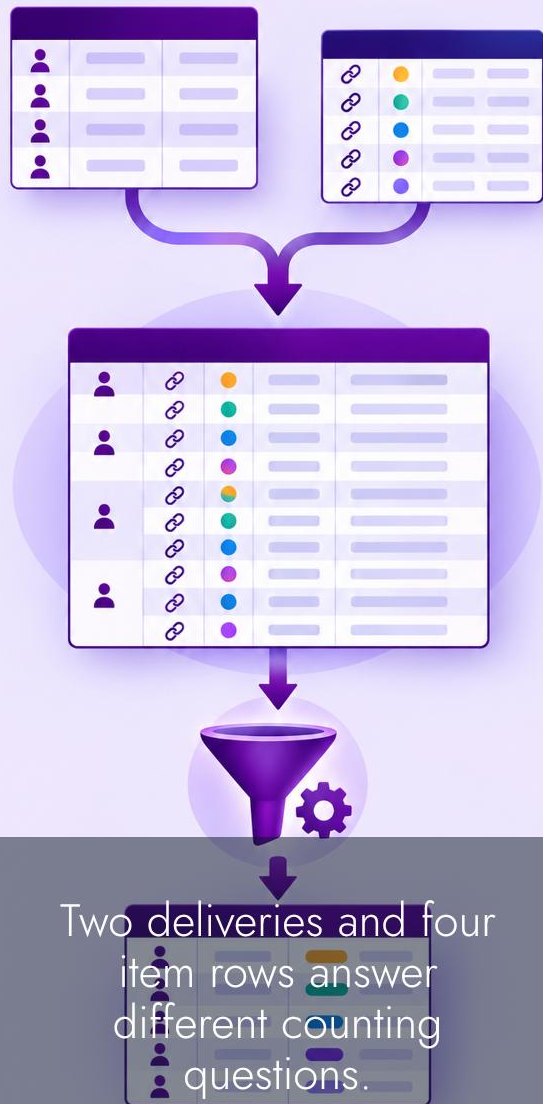
Join Distortion

- Joining four line items repeats Order A three times. The unadjusted table calculates $1 / 4 = 25.0\%$. The deliveries did not change; their row weights changed.

3

Model Repair

- Pre-aggregate items to one row per order before merging. Always verify grain before calculating **COUNT** or **AVERAGE**.



DIAGNOSE THE REPEATED-ROW EXAMPLE

Protect delivery-level meaning before calculating a rate

1

Procedure

- Use the separate two-order illustration on the previous slides. Order A is late and has three items. Order B is on time and has one item.

2

Audit

- Count two deliveries, then four item rows. At delivery grain, the on-time rate is $1 / 2 = 50\%$. Counting item rows incorrectly gives $1 / 4 = 25\%$.

3

Repair

- Keep one observation per delivery before calculating a delivery rate. In Class practice 3, explain why repeating Order A changes its weight without changing either delivery.

AI SEARCH & SHARE



Connect counted units
to a measurable
regulatory consequence

ASK AI: AUDIT A REAL RECORD-GRAIN CASE

Connect counted units to a measurable regulatory consequence

1

Ask AI

- "Using the Wells Fargo enforcement action announced by the Consumer Financial Protection Bureau (**CFPB**) on December 20, 2022, explain why more than 16 million affected accounts cannot be interpreted as
- 16 million customers or incidents. Identify the \$3.7 billion consequence and verify both figures with the dated **CFPB** source outside the **AI** answer.



State one uncertainty
about account,
customer, product, or
incident grain.

ASK AI: AUDIT A REAL RECORD-GRAIN CASE | CONTINUED 2

Connect counted units to a measurable regulatory consequence

1

Ask AI

- State one uncertainty about account, customer, product, or incident grain."

2

Share

- Name the organization, counted unit, verified date and figures, regulatory consequence, and one uncertainty. Do not upload private or restricted data.

RETRIEVAL CHECK

Classify by meaning and protect intended grain

- **Question**

Why is order_id categorical nominal, and how can an Orders-to-Items join distort an order-level on-time rate?

- **Action**

Take 30 seconds to explain field meaning, the one-to-many join, and the 50.0% versus 25.0% rate shift.

- **Next**

Chart Selection and Honest Statistical Critique .



Excel first, SPSS second

CLASS PRACTICE 3: EXPLAIN THE GRAIN

6 minutes in the lab, then back here (slide 55)

1

Open

- Open Session 02 Practice at “Class practice 3: What does one row represent?” Excel cells: **B45, B46, B47:B48, B47, B48, B49**.

2

Do, Excel then SPSS

- Distinguish the 2,000 delivery rows from the separate four-row item example. Classify variables and scales; explain why two orders can produce four item rows.
- **SPSS block:** “Block **H3**: item grain, repeated order IDs, and restoration to order grain”.

3

Bring back

- Bring back a completed type-and-scale classification, two grain statements, and one mixed-grain warning. Timer: 6 minutes.

SECTION

4. CHART SELECTION AND HONEST STATISTICAL CRITIQUE

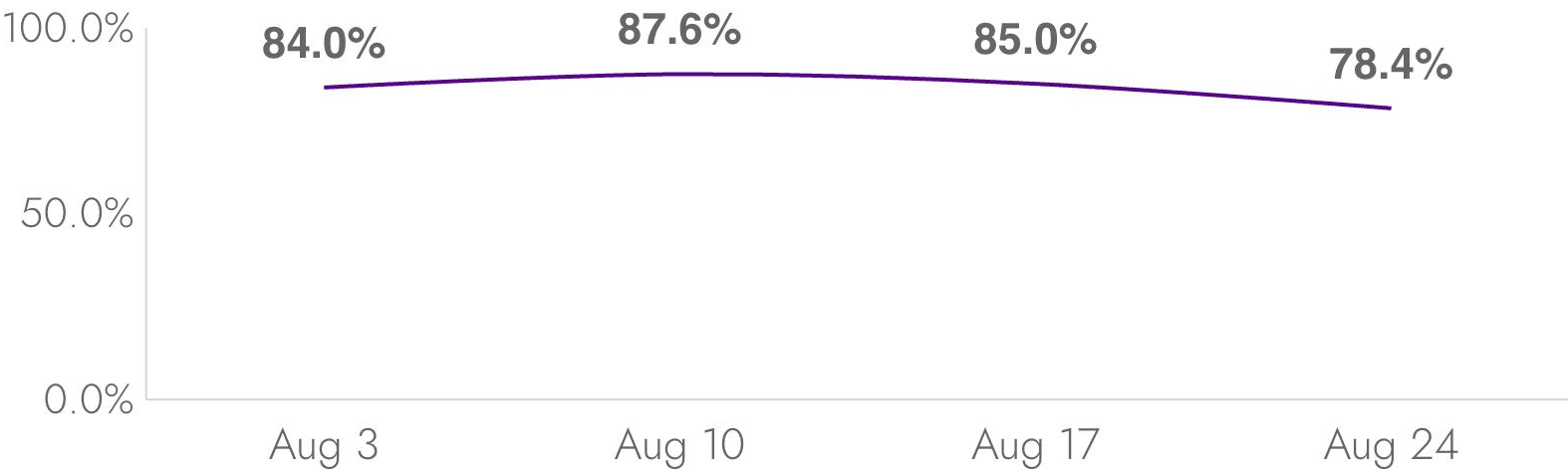
Match the display, scale, and claim to the decision



Tables, bars, and lines support different audience tasks

MATCH THE DISPLAY TO THE QUESTION

Tables, bars, and lines support different audience tasks



The latest rate was 78.4%, below the 85.0% target; four weekly observations show a sequence but do not establish a stable trend or forecast.

500 eligible completed deliveries per week | August 3 through August 30, 2026 |
Unit: percent of eligible completed deliveries



Match visual form to the viewer task.

MATCH THE DISPLAY TO THE QUESTION

Tables, bars, and lines support different audience tasks

1

Definition and rule

- Match visual form to the viewer task. A **table** enables exact lookup, a **bar chart** compares magnitudes across categories, and a **line chart** shows an ordered sequence across time.



Tables, bars, and lines support different audience tasks

MATCH THE DISPLAY TO THE QUESTION | CONTINUED 2

Tables, bars, and lines support different audience tasks

1

Worked example

- MetroMeal's weekly on-time rates are 84.0%, 87.6%, 85.0%, and 78.4% (N = 500 deliveries per week).
- A table audits values, bars compare discrete weeks, and a line highlights the sequence against the 85.0% target.

2

Worked example

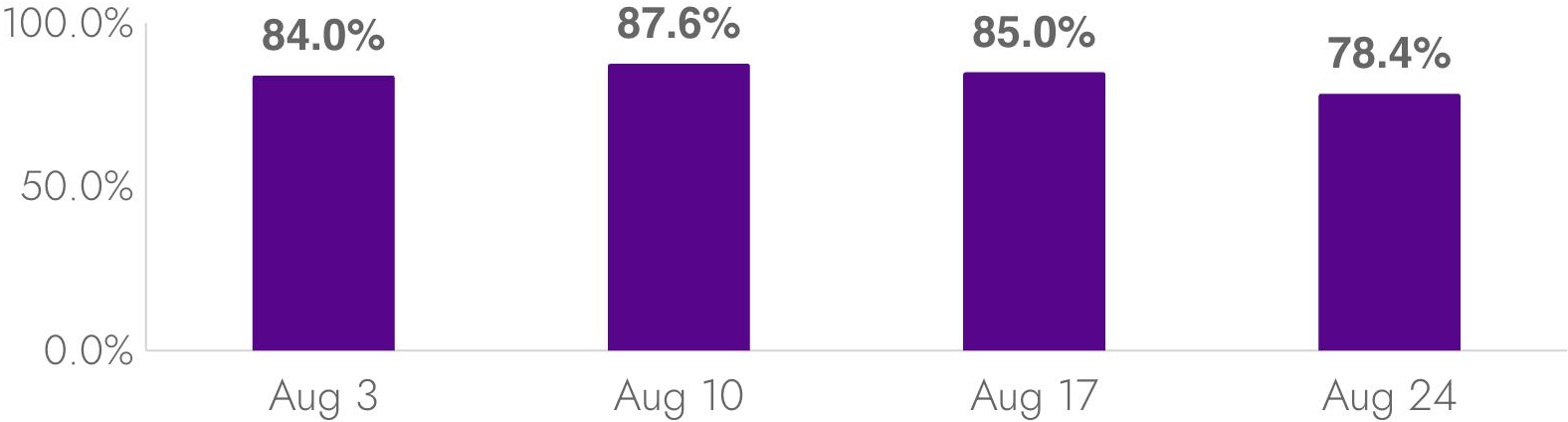
- **⚠ Limit:** Four connected points show an observed sequence, but they do not establish a permanent trend or forecast.



Encoding determines
how viewers judge
magnitude

AXES, BASELINES, AND STATISTICAL MEANING

Encoding determines how viewers judge magnitude



A zero baseline preserves proportional bar length; the latest rate was 78.4%, or 6.6 percentage points below the 85.0% target.

500 eligible completed deliveries per week | August 3 through August 30, 2026 |
Unit: percent of eligible completed deliveries



Encoding determines
how viewers judge
magnitude

AXES, BASELINES, AND STATISTICAL MEANING

Encoding determines how viewers judge magnitude

1

Definition and rule

- An honest axis uses a baseline, range, and scale that preserve visual encoding. Quantitative bars encode magnitude through length, so bar baselines must begin at zero.



Encoding determines
how viewers judge
magnitude

AXES, BASELINES, AND STATISTICAL MEANING | CONTINUED 2

Encoding determines how viewers judge magnitude

1

Worked example

- Truncating MetroMeal's bar axis from 75.0% to 90.0% visually exaggerates the drop to 78.4% into a near-total collapse.
- A zero-to-100% axis preserves the true 6.6 percentage point shortfall against the 85.0% target.

2

Worked example

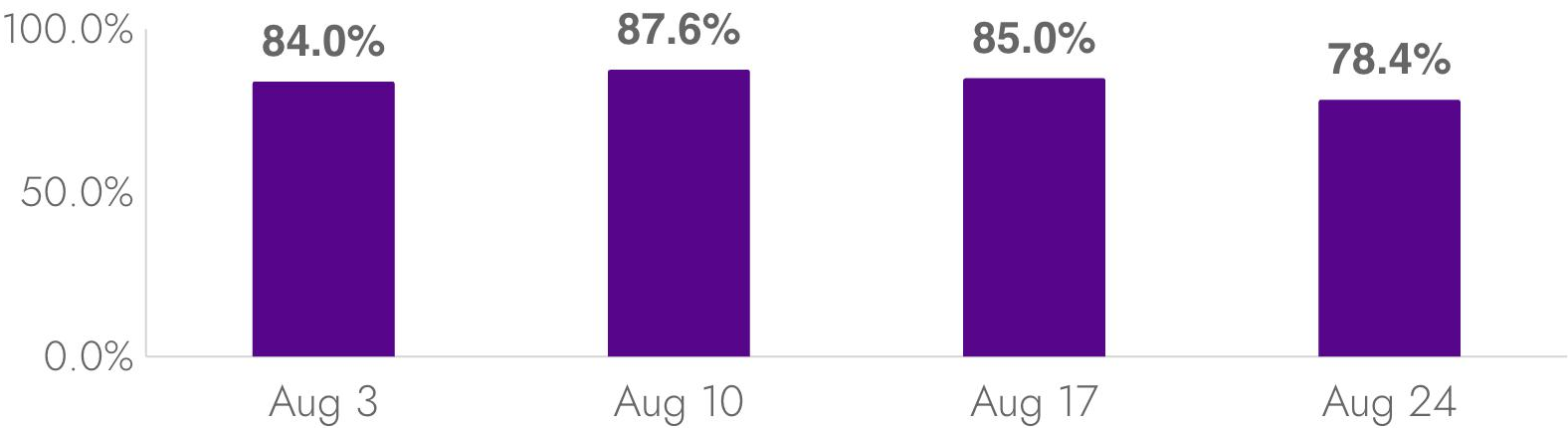
- **⚠ Limit:** Line charts encode position rather than length and may use a narrower labeled range, provided the baseline and practical scale are clearly communicated.



Show denominators,
target, gap, and
non-color cues.

CHART CRITIQUE AND ACCESSIBILITY PROTOCOL

Check meaning and source before appearance



Show denominators, target, gap, and non-color cues.

500 eligible deliveries per week | August 3 to August 30, 2026 | Unit: percent of eligible deliveries



Meaning and source first; encoding and accessibility second; claim and repair last.

CHART CRITIQUE AND ACCESSIBILITY PROTOCOL

Check meaning and source before appearance

1

Honest chart critique

- Check decision, statistic, grain, period, denominator, source, chart type, scale, order, labels, emphasis, clutter, accessibility, title, and limitation.

2

Rule and case

- Meaning and source first; scale and accessibility second; claim and repair last. A truncated 75% bar baseline makes differences look too large.
- Improve the preceding chart by explicitly adding the 85.0% target, weekly n = 500 and latest -6.6 percentage-point target gap.

3

Rule and case

-  **Watch-out:** Correct values cannot repair misleading scale, denominator, title, or context.



Restate MetroMeal's decision and identify the weekly statistic.

APPLY THE CRITIQUE IN EXCEL 365

Trace evidence before changing formatting

1

Procedure

- Restate MetroMeal's decision and identify the weekly statistic. Predict which flaw most affects interpretation.
- Trace every value to the weekly table, verify grain and denominator, then inspect chart type, axis, labels, order, clutter, emphasis, and non-color cues.

2

Finish

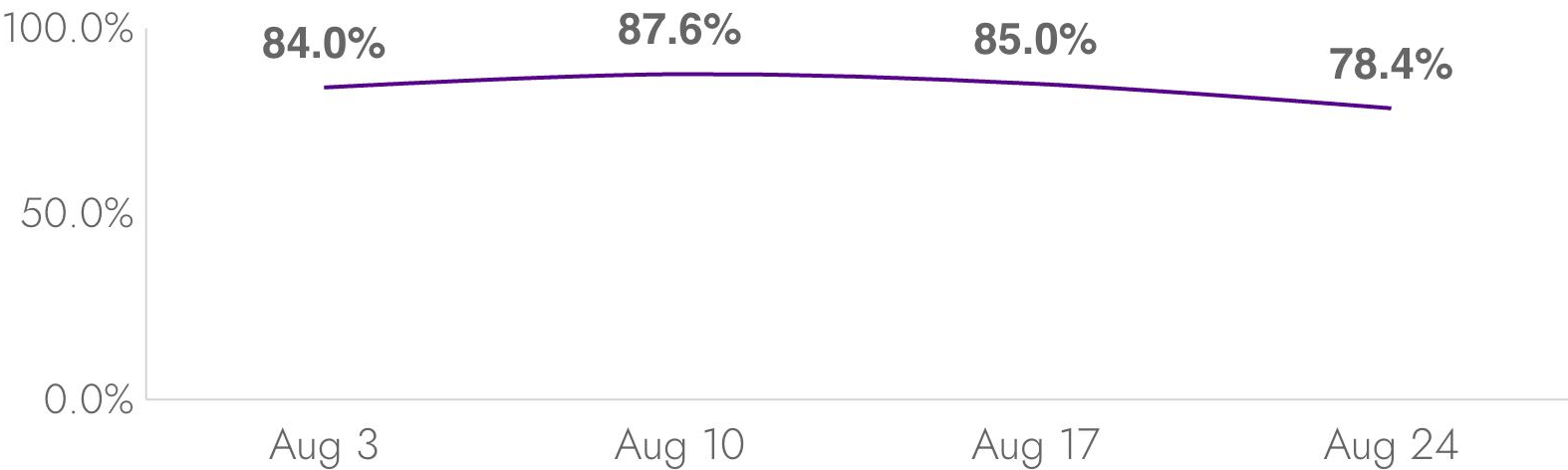
- Test the title against the evidence and name one repair that improves decision usefulness.



State what happened
without inventing cause
or certainty

BOUNDED CHART INTERPRETATION

State what happened without inventing cause or certainty



The August 24 week rate was 78.4%, below the 85.0% target and the prior three observed weeks; four descriptive weeks establish neither cause nor future performance.

500 eligible completed deliveries per week | August 3 through August 30, 2026 |
Unit: percent of eligible completed deliveries



Claim = result +
comparison + scope +
causal and future
boundary.

BOUNDED CHART INTERPRETATION

State what happened without inventing cause or certainty

1

Definition and rule

- A bounded interpretation states the observed measure, population, period, comparison, and evidence limit. Claim = result + comparison + scope + causal and future boundary.

2

Worked example

- From August 24 through August 30, MetroMeal recorded 392 of 500 eligible deliveries on time (78.4%), missing target by 6.6 percentage points and falling below prior weeks.

3

Boundary limit

- Words such as caused, proved, significant, or will require evidence beyond a four-week descriptive record.

AI SEARCH & SHARE



3 minutes to investigate,
2 minutes to share

ASK AI: AUDIT A REAL MISLEADING CHART

3 minutes to investigate, 2 minutes to share

1

Ask AI

- "Find a chart published from 2020 through 2026 by a named company or business news publisher that uses a truncated axis, missing denominator, dual scale, or misleading title.
- Identify the organization, exact values, population, period, distortion, and a measurable consequence involving money, time, customers, transactions, or decision risk.



Do not upload private
or restricted data.

ASK AI: AUDIT A REAL MISLEADING CHART | CONTINUED 2

3 minutes to investigate, 2 minutes to share

1

Ask AI

- Request a dated credible source, then verify the key number outside the **AI** answer."

2

Share

- Name the organization, distortion, verified number and date, consequence, repair, and one uncertainty. Do not upload private or restricted data.

RETRIEVAL CHECK

Explain how encoding changes the baseline rule

- **Question**

Why must MetroMeal's bar chart begin at 0%, while a carefully labeled line chart might use a narrower range?

- **Task**

Take 30 seconds. Explain the difference between length and position encoding, then state the safeguards required for a narrowed line-chart range.

- **Next**

Bounded Story, Validation, and Bridge to Descriptive Evidence .



Excel first, SPSS second

CLASS PRACTICE 4: REVIEW THE EVIDENCE

6 minutes in the lab, then back here (slide 71)

1

Open

- Open Session 02 Practice at “Class practice 4: Does the chart communicate honestly?” Excel cells: **B63, B64, B65, B66**.

2

Do, Excel then SPSS

- Use the lab prompts to critique the 75.0% truncated axis, verify the 78.4% latest rate and 85.0% target, and select an honest repair.
- **SPSS block:** “Block **H4**: four weekly rates and a simple zero-baseline bar chart”.

3

Bring back

- Bring back one statistical defect, one accessibility defect, one repair, and a revised bounded title. Timer: 6 minutes.



Pause. Reset. Return
ready.



10-MINUTE BREAK

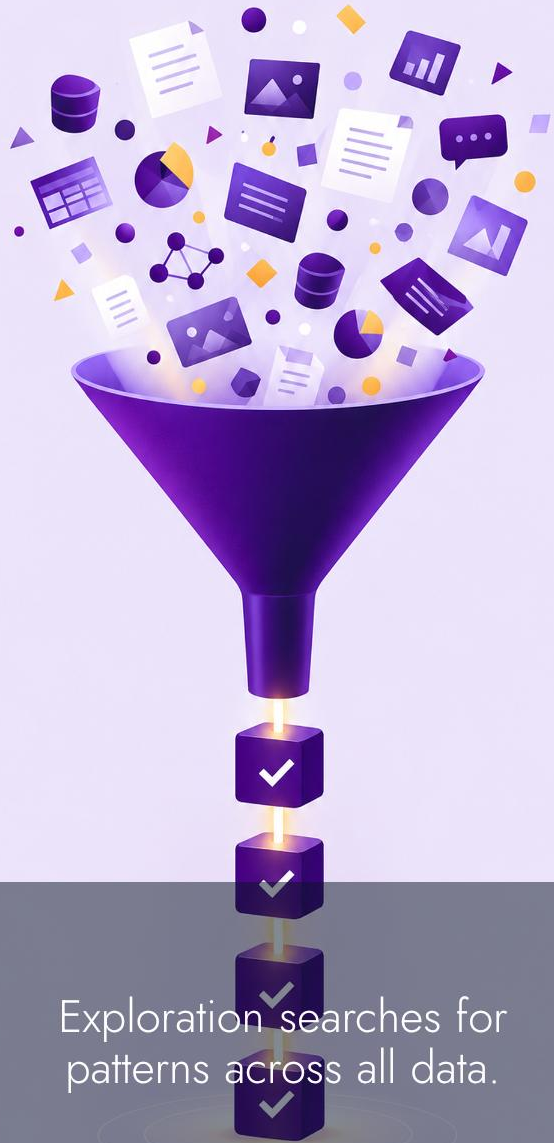
Return for Section 5, then the guided Excel build

Stretch, Hydrate, and Reset

SECTION

5. BOUNDED STORY AND VALIDATION

Turn checked results into a traceable recommendation



Exploration searches for patterns across all data.

FROM EXPLORATION TO A BOUNDED STORY

Select checked evidence for a defined decision

1

Exploration vs Explanation

- Exploration searches for patterns across all data. Explanation selects checked evidence to answer one specific decision question.

2

The Bounded Story Frame

- Combine decision + observed result + benchmark comparison + bounded action + stated limitation + next check.

3

MetroMeal Application

- Week 4 achieved 78.4% on-time delivery (392 of 500 orders), missing the 85.0% target by 6.6 percentage points. This supports investigating dispatch and staffing records, not immediate schedule overhauls.

WHEN IS THE STORY READY?

Decide whether to explain or continue exploring

1

Explain Now

- Share findings when metric definitions, population eligibility, row grain, calculations, chart baselines, and claim boundaries completely agree.

2

Continue Exploring

- Pause release if missing timestamps, unstable filters, duplicate primary keys, or unverified anomalies could change the conclusion.

3

MetroMeal Decision Gate

- Publish the observed target shortfall and recommend a dispatch audit. Hold back any claim attributing delays to driver shortages until verified.



Publish the observed target shortfall and recommend a dispatch audit.



INDEPENDENT VALIDATION AND FORWARD BRIDGE

Challenge the complete chain before recommending action

1

Definition

- Independent validation checks results through source audit, reconciliation, alternative calculation routes, and cross-artifact alignment across tables, charts, and narrative.

2

MetroMeal Checks

- $392 + 108 = 500$ deliveries; $392 / 500 = 78.4\%$; $78.4\% - 85.0\% = -6.6$ percentage points; $14 / 500 = 2.8\%$ complaint rate.

3

Decision Rule

- Stop immediately when two artifacts disagree. Re-running one formula with unchanged inputs verifies arithmetic, not the complete evidence chain.

AI SEARCH & SHARE



Investigate for 3
minutes, then share for
2 minutes

ASK AI: VERIFY A BOUNDED BUSINESS STORY

Investigate for 3 minutes, then share for 2 minutes

1

Ask AI

- "Find a recent annual report from **UPS**, Target, or Starbucks where an operational metric supported a decision.
- Identify the metric, choice, and measurable impact on money, time, volume, customers, or risk. Include the official source, date, limitations, and next validation check."

2

Share

- State the enterprise, verified metric, dated source, decision, measurable consequence, evidence boundaries, and one remaining uncertainty. Verify the key number outside the **AI** answer. Do not upload private data.

RETRIEVAL CHECK

Rebuild the complete traceability chain

- **Question**

Reconstruct the eight-step chain: stakeholder, plausible choice, KPI specification, row grain, calculation, honest chart, bounded recommendation, and next validation check.

- **Task**

Take 45 seconds to map MetroMeal's facts across all eight steps, distinguishing the four-week population from the Week 4 denominator.

- **Next**

Class practice 5, followed by the guided Excel build.



Excel first, SPSS second

CLASS PRACTICE 5: VALIDATE THE STORY

6 minutes in the lab, then the guided build (slide 80)

1

Open

- Open Class practice 5 in the lab guide. Use Slide_Practice cells **B77**: **B82** for the slide case and Class_Practice for the fresh example.

2

Do, Excel then SPSS

- Use the lab prompts to reconcile 392 of 500, 78.4%, -6.6 percentage points, and 14 of 500 complaints, or 2.8%, before writing the bounded story.
- **SPSS block:** "Block **H5**: reconcile the latest evidence and calculate story values".

3

Bring back

- Bring back one explanatory title, a three-sentence bounded story, one action, one limitation, and one next validation check. Timer: 6 minutes.



PRACTICE

Trace a KPI from delivery rows to an honest story



Instructor models, you
build

GUIDED BUILD: TRACE A KPI FROM DATA TO A STORY

15 minutes in the workbook; Self_Check tells you when you are right

1

Watch: inspect the source (4 min)

- **Inspect Orders_Story!A1:P2001**; use Guided_Practice!**B5** and KPI_Build!**B3:B5**. Check 2000, 2000, 2000.

2

Build, chart and validate (11 min)

- **Guided Step 2 (4 min)**: Build weekly **KPI** and guardrail calculations in **KPI_Build!B6:H18**. Check latest row 10 and aggregate row 11.
- **Guided Step 3 (4 min)**: use Chart_Build!**A5:E8** and **A4:C8**. Guided Step 4 (3 min): validate with Self_Check and Response!**B7**.

3

Return

- Return with all completed numerical checks showing ✓ and exactly three sentences in Response!**B7**.



GROUP STUDIO

Repair the *MetroMeal* story



PNP-GRP feedback
studio: repair the
MetroMeal story

GROUP STUDIO: REPAIR THE METROMEAL STORY

10 minutes group work; 5 minutes to share/present per group

1

Roles and file

- Decision and **KPI** checker, Grain and denominator checker, Chart and accessibility critic, Claim and limitation editor. Open the **PNP-GRP** practice document.

2

Use today's lab

- Name the stakeholder and decision before discussing the chart. Trace every quantitative phrase to the **KPI** table or source cells. Check the population, period, grain, numerator, denominator, target, and guardrail.

3

Bring back

- Prepare one ungraded group summary under 200 words: revised title, three-sentence story, bounded action, limitation and next check.
- Then return to synthesis and the **PNP-IND** exit ticket.

KEY TAKEAWAYS

A defensible story is a traceable chain

- **Frame and define**

Start with the stakeholder, decision, action horizon, and required evidence. Define the KPI population, period, grain, eligibility, formula, target, and guardrail.

- **Analyze and display**

Classify variables by meaning. Check grain before joining or aggregating. Match the table, bar, or line chart to the audience task and preserve the denominator.

- **Validate and recommend**

Reconcile counts, calculations, chart labels, and claims. Recommend only the action supported by verified evidence.



Looking ahead to data
quality, distributions,
and descriptive
evidence

THANK YOU

Looking ahead to data quality, distributions, and descriptive evidence

1

Next session

- Data quality, distributions, and descriptive evidence.

2

Next checks

- Provenance, eligibility changes, missing values, duplicate identifiers, unusual delays, spread, and distribution shape.

3

Preparation

- Review Session 03 materials in Brightspace. Bring the listed data file, data dictionary, starter workbook, and checklist. Complete **PNP-PRIMER-01** before class.



Sources supporting
Session 02

REFERENCES

Sources supporting Session 02

1

Recommended references

- Albright and Winston (2020), Business Analytics, 7th edition; Field (2024), Discovering Statistics Using **IBM SPSS** Statistics, 6th edition; Knafllic (2015), Storytelling with Data.

2

Course sources

- **NYU SPS** Fall 2026 approved syllabus for **MASY1-GC1015**; Session 02 Lecture Guide, KPIs, Variables, Grain, Chart Critique, and Bounded Data Stories.

3

Access note

- No textbook purchase is required. The distributed slides and named Brightspace companions contain the required definitions, examples, methods, and limitations.

APPENDIX

APPENDIX A: FOUNDATIONAL KNOWLEDGE

Prerequisites and reinforcement for Session 02

Purpose

Reinforce population, unit of analysis, observation, frequency, relative frequency, and denominator concepts used in Session 02.

Scope

Self-paced review only. This appendix does not introduce Session 03 methods.

How to use

Study each definition, answer the self-check prompts, and compare your reasoning with the session companion.

POPULATION, OBSERVATION, AND UNIT OF ANALYSIS

Define who or what the statement describes



A population contains all eligible entities covered by a statement.

1

Definition

- A **population** contains all eligible entities covered by a statement. The **unit of analysis** is the entity type studied. An **observation** records one unit at table grain.

2

Worked example

- MetroMeal's four-week population contains 2,000 eligible completed deliveries. Each raw observation represents one completed delivery.

3

Limits and check

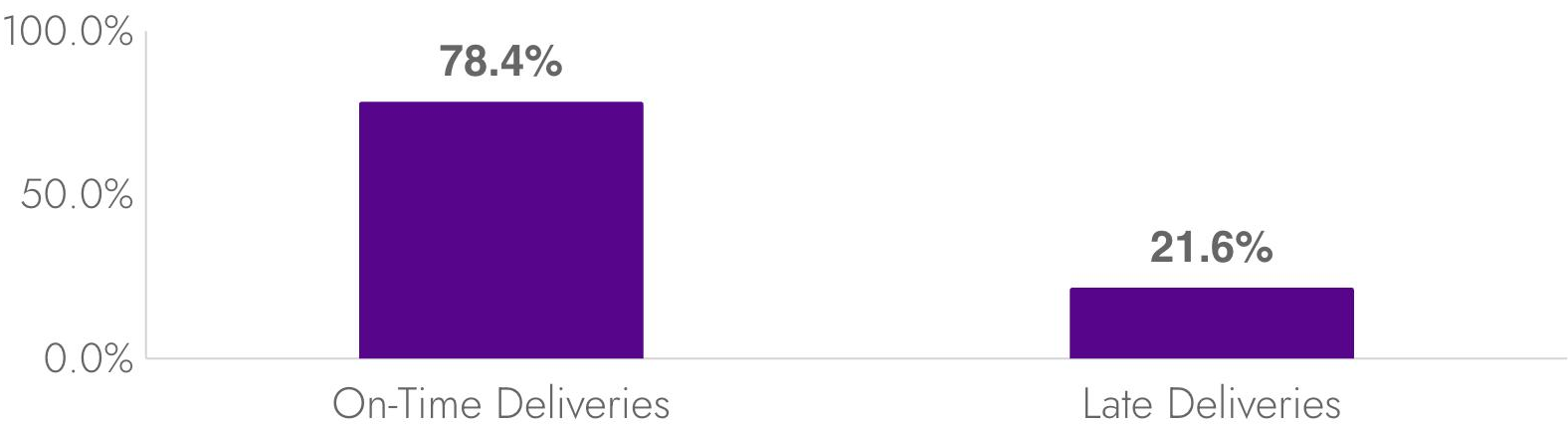
- Customers, couriers, items, and weeks are different units. What exact population does the 78.4% Week 4 rate describe?



Turn a count into an interpretable proportion

FREQUENCY, RELATIVE FREQUENCY, AND DENOMINATOR

Turn a count into an interpretable proportion



392 on-time deliveries divided by 500 eligible deliveries yields a 78.4% on-time rate, while 108 late deliveries represent 21.6%.

completed deliveries in Week 4 | August 24-30, 2026 | Unit: percent of eligible deliveries



Frequency is a count with a unit.

FREQUENCY, RELATIVE FREQUENCY, AND DENOMINATOR

Turn a count into an interpretable proportion

1

Definition and rule

- **Frequency** is a count with a unit. Relative **frequency** divides a qualifying count by the eligible denominator: count / total.

2

Worked example

- MetroMeal recorded 392 on-time and 108 late deliveries: $392 / 500 = 0.784 = 78.4\%$ on-time rate.

3

Limits and check

- A percentage without its denominator hides scale. Ineligible records must never enter the denominator.

APPENDIX

APPENDIX B: EXPLORATION AND STRETCH

Deeper connections, choices, and trade-offs from Session 02

Purpose

Explore deeper choices in KPI governance, weighting, measurement scales, incentives, communication, and claim strength.

Scope

Extend Session 02 concepts without introducing Session 03 methods.

How to use

Examine each trade-off, then ask how it could change a business decision or interpretation.

A KPI AS A METRIC CONTRACT

Version definitions so performance changes remain interpretable

1

Definition and rule

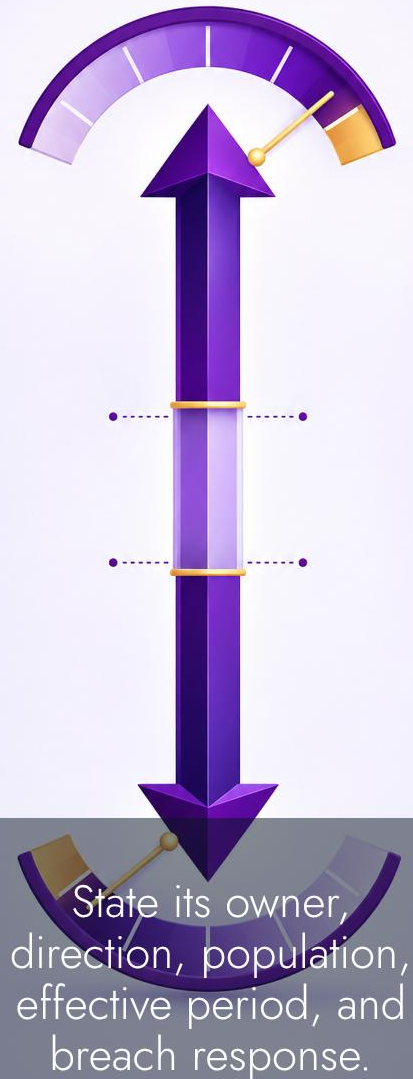
- A metric contract records a **KPI**'s meaning, owner, source, eligibility, calculation, benchmark, refresh schedule, and change history.
- A definition change requires a new version, effective date, reason, approver, and comparison warning.

2

MetroMeal example

- The current contract uses one eligible completed delivery per row, 500 Week 4 deliveries, an 85.0% target, and a 3.0% complaint guardrail.
- **! Check:** Which changes require recalculating earlier results for a fair comparison?

! Check: Which changes require recalculating earlier results for a fair comparison?



TARGETS HAVE DIFFERENT MEANINGS

A benchmark must state its authority, direction, scope, and period

1

Definition and rule

- A target may be a minimum service floor, maximum risk boundary, planning goal, contractual commitment, or temporary threshold. State its owner, direction, population, effective period, and breach response.

2

MetroMeal example

- On-time delivery has an 85.0% minimum target where higher is preferred. Customer complaints have a 3.0% maximum guardrail where lower is preferred.
- **⚠ Check:** Who owns each threshold, and what specific action should a breach trigger?



The correct overall summary depends on the question

DELIVERY-WEIGHTED OR WEEK-WEIGHTED?

The correct overall summary depends on the question

Week	Eligible	On time	Rate
Aug 3	500	420	84.0%
Aug 10	500	438	87.6%
Aug 17	500	425	85.0%
Aug 24	500	392	78.4%

Both summaries equal 83.75% only because each weekly denominator is 500; they answer different questions.

2,000 eligible completed deliveries | August 3 through August 30, 2026 | Unit: eligible deliveries and percent on time



A delivery-weighted rate gives every eligible delivery equal influence.

DELIVERY-WEIGHTED OR WEEK-WEIGHTED?

The correct overall summary depends on the question

1

Definition and rule

- A delivery-weighted rate gives every eligible delivery equal influence. A week-weighted average gives every week equal influence, regardless of delivery volume.

2

MetroMeal example

- $1,675 / 2,000 = 83.75\%$. Also, $(84.0\% + 87.6\% + 85.0\% + 78.4\%) / 4 = 83.75\%$. They match because each week contains exactly 500 deliveries.

3

Check

- Use delivery weighting for the share of all deliveries on time. Use week weighting to describe the average week.

ORDINAL RATINGS NEED AN EXPLICIT CONVENTION

A number code preserves order, not necessarily equal distance

1

Definition and rule

- An ordinal rating from 1 to 5 has a meaningful rank order, but the distance between adjacent rating levels is not guaranteed to be equal.

2

MetroMeal example

- Rating 2 for order 10482 is lower than rating 4. Counts, category percentages, and the median preserve rank order without assuming equal spacing.

3

Check

- A single average rating is compact but hides category distributions. What customer detail is lost when reporting only a mean?



Rating 2 for order 10482 is lower than rating 4.

OUTCOME, DRIVER, AND GUARDRAIL MEASURES

Distinguish results from possible levers



An outcome measure tracks the core business result.

1

Definition and rule

- An outcome measure tracks the core business result. A driver measure monitors a controllable input or process lever. A guardrail protects against unintended negative consequences.

2

MetroMeal example

- On-time rate is the outcome **KPI**. Customer complaint rate is the guardrail. Weekend courier staffing is a plausible driver, but current data do not prove causation.

3

Check

- What additional data or study design could test whether changing courier staffing affects on-time performance?

WHEN THE METRIC BECOMES THE GOAL

Incentives can improve the score while weakening the objective

1

Definition and rule

- Metric gaming occurs when people improve a reported score without improving the underlying objective.
- Goodhart's law warns that a measure can become less useful when strong incentives make it the goal.





Stable rules, visible denominators, audit trails, and complaint guardrails reduce risk.

WHEN THE METRIC BECOMES THE GOAL | CONTINUED 2

Incentives can improve the score while weakening the objective

1

MetroMeal example

- Teams might change eligibility, mark borderline deliveries on time, or rush orders while harming service. Stable rules, visible denominators, audit trails, and complaint guardrails reduce risk.

2

MetroMeal example

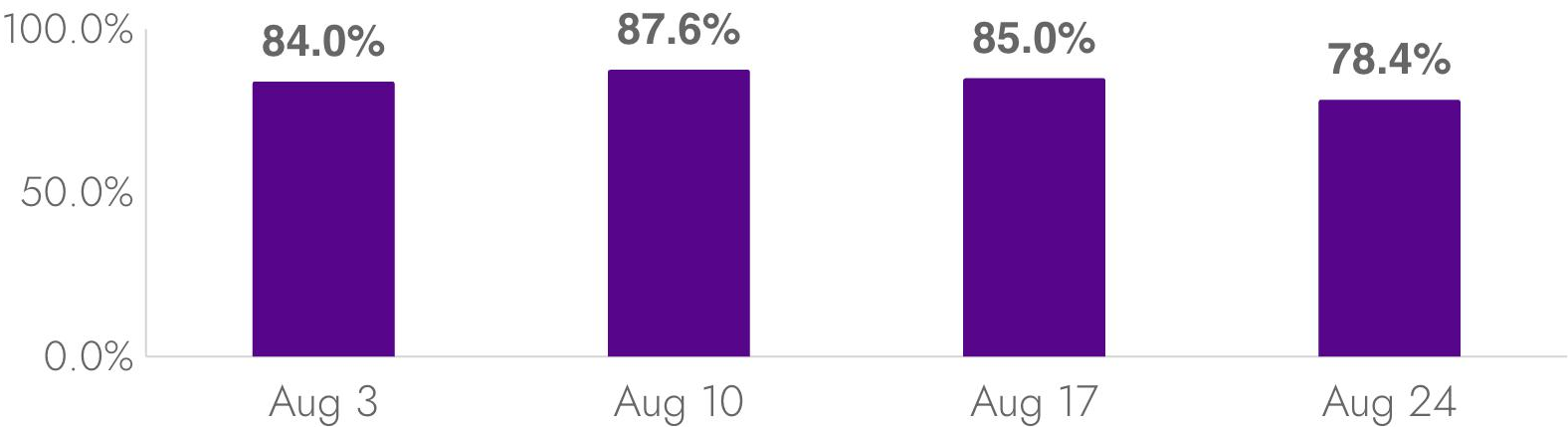
- **⚠ Ethical check:** Who could be excluded, pressured, or harmed by the behavior this target encourages?



Audience task
determines the balance
of precision and
emphasis

ONE DATASET, DIFFERENT COMMUNICATION MODES

Audience task determines the balance of precision and emphasis



Week 4 is lowest at 78.4% and below the 85.0% target; each week contains 500 deliveries, and four weeks do not establish a stable trend.

500 eligible completed deliveries per week | August 3 through August 30, 2026 |
Unit: percent of eligible deliveries on time



A live presentation relies on spoken narrative and visual contrast.

ONE DATASET, DIFFERENT COMMUNICATION MODES

Audience task determines the balance of precision and emphasis

1

Definition and rule

- A live presentation relies on spoken narrative and visual contrast. A circulated document must carry complete definitions, explicit labels, denominators, and boundary limitations without a presenter.



! Check: What context cannot be cut without misrepresenting the analytical claim?

ONE DATASET, DIFFERENT COMMUNICATION MODES | CONTINUED 2

Audience task determines the balance of precision and emphasis

1

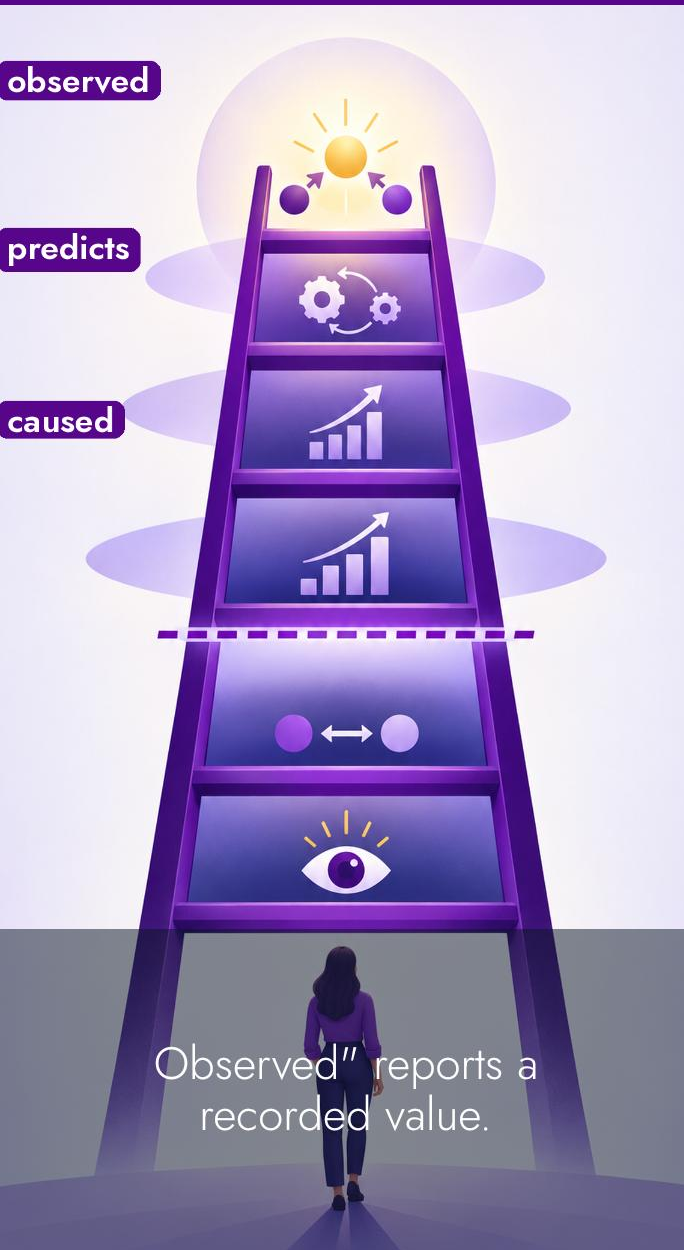
MetroMeal example

- Tables give exact lookup values, zero-based bars show relative magnitude, and lines display sequence.
- Every format must show the 500-order weekly denominator, the 85.0% target, and the August 3 to August 30 window.

2

MetroMeal example

- ! **Check:** What context cannot be cut without misrepresenting the analytical claim?



CLAIM VERBS CARRY EVIDENCE REQUIREMENTS

Stronger wording creates larger analytical obligations

1

Definition and rule

- "Observed" reports a recorded value. "Lower than" makes a descriptive comparison. "Associated with" claims co-occurrence. "Predicts" requires forecasting validation. "Caused" demands evidence that manipulating one variable changed another.



Week 4 reached 78.4%, falling below the 85.0% target and prior weeks.

CLAIM VERBS CARRY EVIDENCE REQUIREMENTS | CONTINUED 2

Stronger wording creates larger analytical obligations

1

MetroMeal boundary

- Week 4 reached 78.4%, falling below the 85.0% target and prior weeks. Staffing can be investigated, but claiming staffing caused the miss or that hiring will fix it is unsupported.

2

MetroMeal boundary

- **! Check:** Underline your strongest recommendation verb. Does your study design actually defend it?

APPENDIX

APPENDIX C: INDUSTRY PRACTICE

Separate reported evidence from teaching inference

Purpose

Examine how major United States organizations define, calculate, communicate, and govern operational metrics.

Evidence boundary

Reported evidence comes from the named organization or regulator. Teaching inference applies course concepts but is not the organization's conclusion.

How to use

Verify each source, identify the metric's unit and population, then evaluate its denominator, grain, comparison, and claim strength.

UNITED STATES DEPARTMENT OF TRANSPORTATION

A low cancellation rate still requires a defined flight population

1

Reported evidence

- On January 3, 2024, the **U.S.** Department of Transportation reported 16.3 million flights in 2023 and a cancellation rate below 1.2%.
- This was the lowest cancellation rate in at least ten years.

2

Teaching inference

- A cancellation rate divides cancelled flights by flights covered by the reporting scope. It describes system performance, not the result of one airline or policy.

3

Governance check

- Confirm flight eligibility, reporting scope, period, exclusions, and revisions before evaluating performance.

A cancellation rate needs a defined flight population and reporting period.



UNITED PARCEL SERVICE

Revenue and package volume are not interchangeable KPIs

1

Reported evidence

- United Parcel Service reported 2024 revenue of \$91.1 billion, 5.7 billion delivered packages, and average daily package volume of approximately 22.4 million.

2

Teaching inference

- Revenue measures financial output in dollars. Package volume measures operational scale in packages. Neither measure establishes on-time performance, service quality, or the cause of change.

3

Governance check

- Align segment, geography, period, operating days, currency, and package eligibility before calculating or comparing measures.

NETFLIX ADVERTISING

Growth depends on a stable definition of the counted unit

1

Reported evidence

- Netflix reported 70 million global monthly active users for its advertising plan on November 12, 2024, compared with 40 million on May 15, 2024.

2

Worked comparison

- $70 \text{ million} - 40 \text{ million} = 30 \text{ million additional users}$. $30 / 40 \times 100\% = 75\% \text{ relative growth}$.

3

Teaching inference and check

- Monthly active users are not accounts, households, or subscribers. Always verify the active-user definition, geography, and measurement window.

70 million - 40 million =
30 million additional
users.

WELLS FARGO AND THE CONSUMER FINANCIAL PROTECTION BUREAU

Control failures can affect millions of distinct accounts

1

Reported evidence

- On December 20, 2022, the **CFPB** ordered Wells Fargo to pay over \$2 billion in redress and a \$1.7 billion civil penalty across 16 million accounts.

2

Teaching inference

- Accounts do not equal unique customers or incidents. Averaging redress across accounts conceals unequal harm across products.

3

Analytical check

- Separate customer grain, account grain, incident counts, consumer redress, and civil penalties.

Accounts do not equal
unique customers or
incidents.